

Research Document (Bike Rental Prediction)

Domain Understanding & Explainable AI

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1. Introduction

1.1 Project Scope and Dutch Context

This research addresses **daily bike-rental forecasting** with a Dutch use case in mind. The Netherlands maintains one of the world's highest cycling shares, with official figures indicating that about **28 percent** of journeys are by bicycle, which makes day-to-day demand management operationally important for service quality and costs. Public bike rentals complement rail travel via **OV-fiets**, available at more than 300 rail-linked locations, reinforcing the need to anticipate daily fluctuations rather than react after the fact.

1.2 Motivation and Practical Value of Daily Forecasting

Daily forecasts help operators plan redistribution and maintenance on the right days, for example reducing unnecessary relocations on low-demand weather days, and improving availability when warm and dry conditions are expected. In cycling cities, day-ahead planning improves reliability and supports sustainability goals.. Guidance for bikeshare operations repeatedly highlights the importance of anticipating demand to guide rebalancing and service levels.

1.3 Document Purpose and Scope

Although the empirical examples later reference the **UCI Bike Sharing Dataset** from Washington, D.C., which provides **daily** rental counts alongside weather and seasonal information, the problem framing is directly relevant to Dutch cities where weather and calendar conditions also drive daily variation. This document establishes the foundation before any modeling, first by clarifying **what Explainable AI means** for a public-mobility forecasting task, second by developing a **domain analysis** that identifies observable **daily** drivers grounded in literature and practice, and third by **deriving data requirements** from the domain rather than from any specific algorithm.

2. Explainable AI, What and Why

2.1 Core Definitions and Key Concepts

Explainable Artificial Intelligence refers to systems whose outputs can be understood by humans, including opaque systems that hide their process, interpretable systems that can be analyzed, and comprehensible systems that present human-readable explanations.

2.2 Importance of Explainability in Public Mobility

Forecasts influence resource allocation, maintenance scheduling, and rider experience, so stakeholders need clarity about the factors shaping each daily prediction. This aligns with European Trustworthy AI principles emphasizing transparency and human oversight across the lifecycle

2.3 Core Principles of Explainable AI

1. **Transparency:** The system should clearly show which factors influence its predictions and how those influences work.
2. **Trustworthiness:** Explanations should be consistent, verifiable, and based on real observations.
3. **Accountability:** Assumptions, limitations, and decision impacts should be documented so outcomes can be reviewed and improved over time.

These principles are widely recognized in both research and European guidance on **Trustworthy AI**.

3. Domain Analysis

3.1 Research Question

Which observable factors influence the daily number of rentals in a city bikeshare system?

This question targets day-to-day variation rather than intraday patterns, because operators plan redistribution and maintenance on a daily cadence.

3.2 Planned Analysis Activities

The analysis draws on academic studies and practitioner materials about cycling and bikeshare demand. It then groups the identified drivers into weather and calendar categories to guide data requirements for a later modeling phase.

3.3 Problem Statement and Rationale

Operators face two recurring risks, empty stations that harm rider experience during peaks, and idle inventory that increases costs when demand drops. In a cycling-centric context like the Netherlands, even modest percentage shifts in daily cycling volumes can have visible operational consequences, which is why anticipating the influence of temperature, rain, wind, and seasonality is valuable for day-ahead planning. Planning guides emphasize that proactive use of demand expectations improves redistribution efficiency and service reliability.

3.4 Observed Factors Influencing Daily Rental Demand

Evidence from Dutch and international studies shows consistent daily relationships between weather and cycling activity, higher temperatures and dry conditions are associated with more trips, while precipitation and strong winds correspond to fewer trips. Calendar structure also matters, seasons and months capture broad climatic cycles, while weekdays differentiate commute patterns from weekend leisure riding. These effects provide a defensible basis for constructing daily features that correspond to observable conditions prior to operations.

3.4.1 Weather-Related Factors

Daily average or maximum **temperature** is positively associated with cycling up to comfort limits, while **precipitation** reduces usage with stronger effects on heavier rain or snow days. **Wind speed** is typically negatively associated with cycling, and **humidity** can interact with temperature and rainfall in shaping comfort. These variables are readily available from standard meteorological sources at a daily resolution, which aligns with the target definition in public datasets.

3.4.2 Calendar-Related Factors

Season and **month** capture climatic cycles and daylight variation relevant to Dutch conditions, while **weekday** separates commute-oriented demand from weekend leisure patterns. **Holidays and large events** can shift volumes away from typical weekday profiles, so daily holiday indicators and event flags are useful where available. These calendar variables are deterministic and easy to generate once the analysis period is defined.

3.5 Summary of Domain Insights

At a daily level, the most consistent observable drivers are weather and calendar factors such as temperature, precipitation, wind, season or month, weekday, and holiday or event indicators. These variables can be measured before operations begin and directly inform redistribution and maintenance decisions.

4. Explainable AI in This Project, Stakeholders and Use Cases

4.1 Operational Perspective (Company Use Case)

Daily forecasts support company teams in planning bike movements, staff allocation, and maintenance activities without reducing availability. Short factor summaries, for example expected temperature, precipitation, wind, season, and weekday, show why tomorrow may differ from a typical peer day and help improve redistribution planning, shift organization, and maintenance scheduling. Add brief what-if lines, for example how stronger rain or calmer wind would change expected demand, so planners can act proactively.

4.2 Customer Perspective (Rider Experience)

Although this project does not directly communicate with riders, improved operational decisions enhance customer experience indirectly. Clear daily forecasts with factor summaries lead to more consistent availability, and later the same summaries could support user messages such as expected high-demand periods or suggested alternative stations.

4.3 Trustworthy and transparent AI applied to decisions in this context

This project does **not** generate automated explanations. Its role is to support planners and operators with **daily forecasts** and simple **factor summaries** such as expected temperature, rain, wind, season, and weekday so they can judge staffing, redistribution, and maintenance needs. At the design stage, the work **aligns** with widely used principles for trustworthy AI being clear about inputs used, documenting assumptions and limits, and keeping humans in the loop for all operational decisions without implementing an explanation module. This keeps the output practical for day-ahead planning while leaving room to add formal explainability later if required.

5. Data Requirements Based On Domain

5.1 Target Definition: Daily Total Rentals

The target variable in this study is the **daily total number of bike rentals** recorded by the system. This approach follows how most public bike-share datasets and operator reports group usage data together with weather and calendar information on a daily basis. Focusing on daily totals keeps the prediction timeline aligned with real operational decisions, such as planning bike redistribution and scheduling maintenance for the next day.

5.2 Input Variables and Optional Operational Signals

Daily weather features such as temperature, rainfall, wind speed, and humidity are key factors that affect cycling activity and are available from national weather services. Calendar features like season, month, and weekday capture climate cycles and travel habits, while holiday or event flags explain unusual days. Optional operational data, such as available bikes, docks, or maintenance times, can help separate real demand changes from supply limits when available.

5.3 Granularity and Time Span

All inputs should be **aggregated or sourced at daily granularity** to match the target. The training period should cover **at least one full annual cycle** to expose the model and explanations to seasonal variation. If possible, include two years to improve robustness of seasonality estimates and to support simple out-of-time validation at the daily level.

5.4 Data Quality and Governance

Define **units and encoding** up front, for example temperature in degrees Celsius, precipitation in millimeters per day, wind speed in meters per second, and calendar variables as categorical features. Specify a **missing-value policy** for weather and target fields, for example drop days with missing target.

5.5 Candidate Data Sources

- **Bike-share usage**, the UCI **Bike Sharing** dataset from Washington, D.C., provides daily rental counts together with weather and seasonal information and serves as a well documented public exemplar for field names and daily structure during the Design phase.
- **Dutch weather**, the **KNMI** open data platform exposes national observations and documentation for station data and daily variables that can supply temperature, precipitation, wind, and humidity at the required resolution.
- **Service context**, operator and planning materials illustrate how daily forecasts inform redistribution and service levels, these sources help align variable choices and explanation formats with real operational decisions in the Netherlands and abroad.

6. Research Methodology: DOT Frameworky

This study follows the Development Oriented Triangulation (DOT) Framework through two strategies. I conducted library research to review academic and practitioner sources on bike-rental forecasting and Explainable AI, ensuring a solid theoretical foundation. I also carried out a community scan to gather insights from developer and operator discussions, identifying real operational challenges and practical variables. This combination ensures that the work remains both evidence-based and grounded in practice.

7. References and Resources

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